

Unsupervised classification of airborne laser scanning data to locate potential wildlife habitats for forest management planning

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To account for ecological objectives in forest management planning, potential habitats need to be mapped, characterized and evaluated for utility in alternative management practices. Airborne laser scanning (ALS) is increasingly used to derive predictive maps of habitat quality. Unlike ecologically driven approaches that require spatially and temporally co-located training data of the specific species, we tested whether indicative information on the habitat potential could be obtained by means of an unsupervised classification of ALS data. Based on a literature review, altogether five ALS features quantifying vegetation height, cover and diversity were expected to capture the essential variation in the habitat requirements of western capercaillie (*Tetrao urogallus* L.) and hazel grouse (*Tetrao bonasia* L.), which are the most important game birds occurring in the studied area. The features were extracted from sparse density, leaf-off ALS data at a resolution of 256 m² and partitioned using an unsupervised *k*-means algorithm. By analysing the persistence of the cluster ensemble formed by the partitioning, altogether 158 plots in 16 structural classes were assigned for field measurements to determine which real-world forest phenomena affected the clustering. The clustering was found to stratify the area mainly in terms of size-related attributes such as timber volume and basal area. The understorey, shrub and herb layers had less correspondence with the clustering, indicating that an unsupervised classification is not directly suitable for habitat mapping. The result was improved using empirical threshold values for the ALS features determined according to the plots labelled as the most potential habitats in the field measurements. This semi-supervised classification of the data indicated 4 per cent of the total forest area as suitable for the specific species, which appears a reasonable estimate of the core area of the habitats considered. Overall, the partitioning formed aggregated, stand-like spatial patterns, even though the neighbourhoods of the individual 256 m² cells were not considered at all. The result could be further refined by spatial optimization to produce indicative maps for forest management planning with ALS as the sole data source.

Keywords: biodiversity, habitat suitability, remote sensing, light detection and ranging (LiDAR), isocluster, data mining

Introduction

Forest management planning often involves numerous decision makers or other stakeholders with different objectives concerning the use of the forest (Kangas *et al.*, 2008). In addition to conventional timber production, the management objectives may be related to maintaining recreational and aesthetic opportunities, historical features, or habitats for wildlife or other organisms. Multi-functional comparisons are needed to evaluate the management alternatives and to determine an optimal choice of the treatments according to the stakeholder's preferences and the production possibilities of the planning area (e.g. Kurttila *et al.*, 2001; Kurttila and Pukkala, 2003; Laiho *et al.*, 2011; Pukkala *et al.*, 2011).

In Finland, for example, the forest legislation was reformed in 2014 to allow more flexible management practices. Consequently,

specific management recommendations were developed for wildlife or game management (Lindén *et al.*, 2014). The locations and properties of potential wildlife habitats are, thus, important information for improved forest planning decisions. In the presence of adequate information, the habitat suitability can be modelled (Kangas *et al.*, 1993) and incorporated as an objective on choice of treatments in numerical optimization (Kurttila *et al.*, 2002). However, obtaining information on habitat quality is time consuming and expensive based on conventional field measurements and sampling.

Recent developments of remote sensing technologies have enabled an accurate three-dimensional mapping of forest structure and subsequent habitat quality assessments for various species and vast areas. The use of airborne laser scanning (ALS; also known in some instances as airborne scanning LiDAR) in

particular has matured and become increasingly popular for various forest inventories (Maltamo *et al.*, 2014). The descriptions of forest structure derived from the ALS data (e.g. Zimble *et al.*, 2003; Frazer *et al.*, 2005; Maltamo *et al.*, 2005; Korhonen *et al.*, 2011) have the potential to assess foliage height diversity, per cent vegetation cover and horizontal patchiness, which have long been recognized as key components of the biological diversity in forests (MacArthur and MacArthur, 1961; MacArthur *et al.*, 1962; Karr and Roth, 1971; Roth, 1976; James and Wamer, 1982). Even more complex structural indices may be developed from the ALS data based on a detailed ecological rationale (Listopad *et al.*, 2015) or a thorough understanding of the properties affecting the ALS response at various scales (Valbuena *et al.*, 2013, 2014). Using the predictor features and wildlife distribution patterns observed from field plots, count stations or pitfall traps as training data for a supervised learning method, predictive maps of habitat quality for certain species may be derived (e.g. Hill *et al.*, 2004; Seavy *et al.*, 2009; Goetz *et al.*, 2010; Smart *et al.*, 2012). The use of ALS and field data for the aforementioned purpose has recently been reviewed in detail by Davies and Asner (2014), Hill *et al.* (2014), Müller and Vierling (2014) and Simonson *et al.* (2014).

'Supervised' learning approaches described above require a considerable care in obtaining a representative set of observations for the training of an adopted model structure. In contrast, 'unsupervised' learning approaches aim at a data-driven clustering, where the available data are partitioned into sensible clusters for deriving the conclusions. The latter approach inherently requires the existence of distinct groups corresponding to the real-world phenomenon of interest in the auxiliary data. In the absence of clear clusters, the performance of an unsupervised method could potentially be improved by a mixture approach called 'semi-supervised' learning (e.g. Theodoridis and Koutroumbas, 2009, p. 577), in which both unlabelled data and a small sample of labelled data are jointly analysed.

Earlier ALS applications of unsupervised methods have related to mapping built environments (e.g. Priestnall *et al.*, 2000; Kong *et al.*, 2013), tree segmentation (Morsdorf *et al.*, 2004; Gupta *et al.*, 2010) and tree species recognition (Reitberger *et al.*, 2008; Yao *et al.*, 2012). Touching on the topic of this study, Pascual *et al.* (2008) segmented ALS data collected from heterogeneous forest structures and classified the obtained stands into five groups based on an unsupervised *k*-means approach. According to a control inventory, the groups corresponded well with the structural differences observed in the field. Apart from this inspiring example, area-based ALS applications of unsupervised learning are most often related to refining the predictor space for later analyses based on supervised methods (e.g. Zhang *et al.*, 2011).

In conclusion, unsupervised or semi-supervised learning techniques could potentially be employed to reduce the efforts in data collection and model fitting required by supervised methods when appropriate features corresponding to species-specific habitat requirements exist in the auxiliary data. In Finland, the most important game birds, for which habitat-preserving silvicultural management is instructed (Lindén *et al.*, 2014), are western capercaillie (*Tetrao urogallus* L.) and hazel grouse (*Tetrastes bonasia* L.). The habitat requirements of both the species are characterized by distinct classes in terms of the forest vertical and horizontal structure and successful examples of modelling of these requirements by ALS exist (Breidenbach and Braunsch, 2008; Graf *et al.*, 2009; Zellweger *et al.*, 2013, 2014). Even though the locations of

the most important areas to these species, such as lekking sites of *T. urogallus* (e.g. Sirkiä *et al.*, 2012), are well known, complete spatially explicit information that would support prioritizing areas for wildlife management within forest management planning is not provided by regular forest inventories.

ALS data are nowadays routinely collected for ground elevation modelling of extensive geographical areas and several studies have also verified the suitability of the data for various forest inventories (Nord-Larsen and Riis-Nielsen, 2010; Nord-Larsen and Schumacher, 2012; Villikka *et al.*, 2012; Kankare *et al.*, 2015; Vauhkonen *et al.*, 2015). The vertical and horizontal distribution of the ALS data may be condensed to a few metrics corresponding closely to the structural complexity observed in the field (Kane *et al.*, 2010; Lehterä *et al.*, 2015), i.e. a stratification of the area per forest structure, which is useful *a priori* information for several inventory applications (Dalponte *et al.*, 2011; Maltamo *et al.*, 2011; Gobakken *et al.*, 2013). We further hypothesize that by identifying metrics that reflect the habitat requirements, unsupervised or semi-supervised approaches could provide an *a priori* map on the potential habitats, reducing the needs for species-specific field training data required by the supervised predictions and, consequently, problems related to co-locating these observations in the spatial and temporal dimensions (Vierling *et al.*, 2014).

The purpose of this study was to examine the results of an unsupervised classification of a forested area based on structural metrics extracted from ALS data. The ALS metrics were selected to depict the habitat potential for *T. urogallus* and *Tetrastes bonasia* according to a literature review on the requirements of these species. A field survey was designed to sample the forest attributes corresponding to the obtained clustering. Options to use the obtained information for habitat management or forest management planning in general are discussed.

Materials and methods

Study area and data

The study area is located in the southern boreal forest zone (61.19° N, 25.11° E). The area of altogether ~2000 ha is a part of a state-owned forest. The forest stands vary from natural to intensively managed in terms of their silvicultural status. The area is dominated by Scots pine (*Pinus sylvestris* L.) and Norway spruce (*Picea abies* [L.] H. Karst.), which constitute ~80 per cent of the growing tree stock according to earlier studies on the same area (e.g. Kankare *et al.*, 2015; Vauhkonen *et al.*, 2015). Minor proportions of deciduous species such as birches (*Betula* spp. L.), aspen (*Populus tremula* L.), alders (*Alnus* spp. P. Mill.), willows (*Salix* spp. L.) and rowan (*Sorbus aucuparia* L.) typically occur below the dominant canopy.

The ALS data used in the study were acquired by the National Land Survey of Finland as a part of their data acquisition campaign for creating a nationwide ground elevation model for Finland. The data were downloaded from a file service (National Land Survey of Finland, 2015), from which they are available for free and with extensive permissions of use. The data were acquired in two separate campaigns: on 7 May 2012, with Optech ALTM Gemini scanner operated from a flying altitude of 1830 m, and on 13 May 2012, with Leica ALS50 scanner operated from 2200 m. Similar scanning parameters were applied in both the acquisitions to yield a nominal pulse density of 0.7–0.8 m⁻². The data provider had detected and classified the ground level, on which the normalization of the vegetation height values was based. As the data are meant specifically for ground elevation modelling, we assumed the accuracy of this classification to be appropriate for our purposes.

The data were tessellated to a regular grid with a cell size of 16 × 16 m corresponding to general area-based ALS analyses (e.g. [Næsset, 2014](#)). All non-forest areas such as lakes, buildings, roads and areas within 20 m of these objects were removed from the analyses based on an existing forest management planning map. Finally, mature forests with more potential for the habitat selection (see the next section) were distinguished from the younger development stages and retained for the analyses employing a threshold of 10 m for the minimum allowable vegetation top height. Altogether 46 201 (56 per cent) of the initial 81 862 grid cells for the entire area were retained for the analyses.

ALS feature selection according to the habitat requirements

Habitat requirements for the two species were extracted from an earlier literature review ([Imponen, 2015](#)). Both of the birds considered are ground-nesting species, for which reason an adequate cover of herbaceous understorey vegetation, trees and shrubs are essential habitat criteria to provide nutrition, shelter from the weather and protection against predators. Both the species also benefit from heterogeneity of the vertical canopy, tree species and size distributions, while the specific requirements vary per species. *Tetrastes bonasia* tends to prefer very dense, mixed stands typically occurring in 20–50 years after a disturbance such as forest fire. The natural habitats of *T. urogallus* are old (>90 years) pine-dominated forests with a rich interior structure due to a frequent disturbance history. The habitats of *T. urogallus* are generally characterized by low tree density and a light canopy, but the presence of more densely covered patches increases the preference. Even if the habitat requirements were described above in terms of stages related to the natural forest succession ([Swenson and Angelstam, 1993](#)), the species also dwell in managed forests, favouring similar structural properties ([Åberg et al., 2003](#); [Miettinen et al., 2010](#)). A characterization of these properties is summarized in Table 1.

Accordingly, we selected the ALS variables based on three main criteria:

- (1) They have to be a proxy for habitat requirements identified during field surveys.
- (2) They have to be obtainable from the sparse density data.
- (3) Selected variables should have low inter-correlation to minimize redundancy of the information and thereby ensure an equal weight of each variable (serving as a proxy for one habitat requirement) in the unsupervised classification.

Since no information on the species composition of the herbaceous understorey vegetation (hereafter ‘herb layer’) was expected to be obtainable by ALS, the features were selected to describe (i) the height and (ii) canopy cover of the dominant trees, (iii) the density and (iv) diversity of the understorey trees and (v) the proportion of the shrub vegetation. Applying thresholds of 0.5 and 5 m for the ground and canopy, respectively, and considering the observations in between as the shrub layer, the aforementioned features were calculated per grid cell as:

- (i) the average height of the highest 10 per cent echoes (denoted in the following text as H_{max});
- (ii) the ratio of the echoes reflected from the canopy to all echoes, i.e. canopy cover (CC);
- (iii) the ratio of the echoes reflected above ground but below 60 per cent of the maximum height to all echoes reflected above ground (DEN_{60});
- (iv) the standard deviation of the height values of echoes reflected above ground but below 60 per cent of the maximum height (SD_{60}) and
- (v) the ratio of the echoes reflected from the shrub layer to all echoes (US).

The selection and definitions of the features were based on the habitat requirements (Table 1), earlier ALS analyses carried out in Central Europe ([Breidenbach and Braunisch, 2008](#); [Graf et al., 2009](#); [Zellweger et al., 2013, 2014](#)) and experimenting with the local data. In young forests, features (iii)–(v) may describe partly similar properties. However, the selected metrics are most likely a good compromise in obtaining separate descriptions of the vegetation structure in the shrub layer (v) and between the shrub layer and dominant canopy (iii, iv) in mature forests and with respect to the most important habitat characteristics (cf. Table 1). Excluding (ii), which was computed using only the first echoes (i.e. ‘only’ and ‘first of many’ of up to four echoes recorded per pulse; see [Korhonen et al., 2011](#)), the metrics were based on all echoes recorded. The Pearson correlations between the features were mainly low (Table 2).

Unsupervised classification

To include all the ALS features with the same importance in the clustering, the original feature values were normalized between 0 and 1. The normalized feature values were obtained as

$$r_{ij} = \frac{q_{ij} - q_i^{\min}}{q_i^{\max} - q_i^{\min}}, \tag{1}$$

where r_{ij} and q_{ij} are the normalized and original values of the j th observation of feature i , and $q_i^{\min}$ and $q_i^{\max}$ are the smallest and largest values of i among all grid cells. The normalized feature values, thus, covered the full range of the ALS features within the grid.

The data were clustered using a k -means algorithm ([Hartigan and Wong, 1979](#)) implemented in R ([R Development Core Team, 2013](#)). The k -means method partitions the observations into k clusters such that the sum of squares from the observations to the assigned cluster centres is minimized. Due to the low inter-correlation of the features considered (Table 2), the clustering was based on the standard Euclidean distance. Altogether 10 000 random solutions were used as initial cluster centres to minimize the randomness in the final result.

Since it was found problematic to determine a fixed k , i.e. to fix the expected value of the structural classes, we determined the final partitioning according to the persistence of the clusters on an interval of different k values. Based on initial tests in the data studied, we expected that

Table 1 A summary of the structural properties of the habitats preferred by the species considered

Habitat characteristics ^a	Western capercaillie ¹ (<i>Tetrao urogallus</i> L.)	Hazel grouse ² (<i>Tetrastes bonasia</i> L.)
Dominant species	Pine; spruce in peatlands in the summer time	Spruce with a deciduous mixture
Vertical strata	Two or more	Multiple
Canopy cover	Average (<70%), with heterogeneity corresponding to a natural state	High (60%–80%) with canopy gaps
Understorey cover	Average; patchy	High
Species in the herb layer	<i>Vaccinium vitis-idaea</i> , <i>V. myrtillus</i> , <i>Melampyrum</i> spp.	<i>V. myrtillus</i> (in average-high density), <i>Oxalis acetosella</i> , <i>Melampyrum</i> spp.

^aBased on [Angelstam et al. \(2004\)](#)^{1,2}, [Swenson and Angelstam \(1993\)](#)^{1,2}, [Miettinen et al. \(2010\)](#)¹ and [Uusvaara \(1963\)](#)².

Table 2 The correlations of the features selected

	$H_{\max}$	CC	US	DEN ₆₀	SD ₆₀
$H_{\max}$	1	0.571	-0.596	-0.070	0.796
CC	0.571	1	-0.770	-0.411	0.550
US	-0.596	-0.770	1	0.583	-0.633
DEN ₆₀	-0.070	-0.411	0.583	1	-0.352
SD ₆₀	0.796	0.550	-0.633	-0.352	1

$H_{\max}$, the average height of the highest 10% echoes; CC, the ratio of the echoes reflected from the canopy to all echoes; DEN₆₀, the ratio of the echoes reflected above ground but below 60% of the maximum height to all echoes reflected above ground; SD₆₀, the standard deviation of the height values of echoes reflected above ground but below 60% of the maximum height; US, the ratio of the echoes reflected from the shrub layer to all echoes.

meaningful structural information was obtained with k between 4 and 7, this choice being further reasoned in 'Discussion and conclusions'. The clustering was, thus, repeated by gradually increasing the value of k from 4 to 7, using the R package 'clue' (Hornik, 2005) to manage the resulting ensemble of the k -means partitions. During the iterations, it was recorded whether the individual cells persisted in a cluster or shifted from a cluster to another one. As a result, each grid cell was labelled with an identifier describing its path along the clustering hierarchy. Therefore, as opposed to results with a fixed k , the final partitioning according to these labels provided a means to analyse the composition of the sub-clusters as these were formed during the clustering.

Sampling of the partitioned data for validation

Based on analysing the initial results, the individual class labels with a number of cells <1 per cent of the total number were deemed as clustering noise and thus unimportant with respect to the studied phenomena. The cells within each remaining label were ordered based on a product $H_{\max} \times CC$, and the cells corresponding to the 10, 25, 50, 75 and 90 percentiles of this product were assigned for field measurement. The purpose of this step was to divide the sample to cover the range of the size and density of growing stock within each label, for which these features correspond well (e.g. Maltamo et al., 2011). After obtaining the initial observations from each label with a frequency ≥ 1 per cent of the total number of cells, the remaining measurements were focussed on those labels that seemed most interesting in terms of the habitat characteristics of the species studied as observed during the inventory. This was achieved by measuring cells corresponding to intermediate percentiles around the aforementioned ones.

Field measurements

The field measurements were carried out during June – August, 2014. A field measurement protocol based on circular 9 m plots, corresponding to the size of the grid cells, was established. General stand attributes, those related to the shrub and herb layers, which were supposed to depict the habitat requirements of the game birds considered, and tree attributes from trees with a diameter at breast height (DBH) ≥ 5 cm were measured (Table 3). For each species of the plot, a tree with a DBH corresponding to the basal area weighted mean diameter (D_{GM}) was determined in the field and measured for height (H_{GM}). The similarity of the close surroundings of each plot to the 9 m area measured was quantified by establishing an additional plot 16 m from the original plot centre in each cardinal direction (0, 90, 180 and 360°). In these satellite plots, only the trees were tallied employing a basal area factor of 2 m² ha⁻¹.

Finally, the 9-m plots were graded in the field for their overall habitat suitability in terms of the forest structure, and plots resembling the potential habitats of *T. urogallus* and *Tetrastes bonasia* were separately marked.

Table 3 The measurement intervals and resolutions of the attributes measured in the field

Attribute	Measurement interval	Resolution
Tree species	Individual species of the area	Species
DBH	≥ 5 cm	± 0.05 cm
Height	0.5 m – 5 m (shrub layer) >5 m (dominant trees)	± 2.5 cm ± 0.25 m
Age	>5 years	1 year
Number of vertical strata	Dominant, intermediate, understory	1
Number of spruces with low branches (SLR)	Spruces with branches reaching ground level	1
Blueberry and vegetation cover in the herb layer	0%–100%	$\pm 2.5\%$
Grade	0–10	1

The grade given on a scale of 0–10 combined ocular estimates of the canopy cover, the number and heterogeneity of the vertical strata, the quantity and quality of the understory, the amount of blueberry (*Vaccinium myrtillus* L.) in the herb layer and the number of shelter trees with low branches with respect to the habitat requirements. In a coniferous forest, the presence of deciduous trees increased the grade. Thus, even if the grade was based on the characteristics assessed by the field survey, it was a score based on expert opinion by a surveyor with experience of hunting the game bird species.

The plot centres were positioned with a Trimble® Nomad® 900G Global Positioning System and Global Navigation Satellite System receiver with an external antenna and battery. At least 100 positions were collected from the centre of each plot. The receiver was connected to a GSM phone, which acted as a reference station for real-time differential correction to help the positioning in the field. A differential post-processing was finally applied using Trimble® Pathfinder® Office and a local, permanent base station. The accuracy of locating the grid cells was verified by comparing the Euclidean distances between the planned and observed plot positions, i.e. the centre points of the grid cells and the final points located in the field, respectively.

Plot attributes derived for the evaluation

The results of the unsupervised classification were characterized using the plot attributes compiled and aggregated from the measurements (Table 3). Plot basal area (G) was computed by summing from the DBH measurements. The DBH and height corresponding to the basal area weighted mean tree over all species were determined according to the DBH

measurements and used as the D_{GM} and H_{GM} , respectively, of the entire plot. The variations in the within-plot DBH distribution and G observed from the 9-m plot and the satellite plots were quantified using the coefficient of variation (CV), i.e. the ratio of the standard deviation to the mean, of the specific values. The missing tree heights were predicted by calibrating the prediction models for the parameters of Nöslund's (1936) height curve presented by Siipilehto (1999) using the species-specific D_{GM} and H_{GM} estimates. The volumes of the individual trees were predicted by models of Laasasenaho (1982), employing the DBH, height and tree species as predictors. The models for birch were used for all deciduous trees. The volumes (V) were summed to the plot level and scaled per hectare.

Evaluation

We first examined the distributions of the ALS feature values and the locations of the sampled plots in the sub-clusters over the entire grid to determine whether the applied sampling strategy efficiently spread the sample across the studied phenomenon. Second, as it was motivated in the 'Introduction' that an unsupervised clustering might not as such result in classes corresponding directly to the habitats, we analysed the field measurements in two ways:

- (i) We determined whether the unsupervised clustering directly resulted in classes describing the considered habitats. To achieve this, we compared the clustering with the field measurements and attempted to characterize the result according to the forest type and habitat-specific properties.
- (ii) We determined whether the unsupervised result could be improved by labelling a sub-sample according to alternative classification paradigms such as supervised or semi-supervised learning. We determined which of the measured plots constituted the most potential habitats in the field data by comparing the grades and forest attributes with the habitat requirements reported in the literature. We then used these plots as training data to learn the properties of the most potential habitats in terms of the ALS features. In practice, we determined empirical threshold values for the five ALS features, according to which all the plots identified as the most potential habitats could be selected based on the ALS data. We assessed the omission and commission error rates when applying these threshold values to the data. This analysis was made to generalize the results over the entire area and no separate validation was carried out.

Results

The partitioning of the 46 201 grid cells to 4–7 clusters resulted in 43 sub-clusters in the clustering hierarchy. Altogether 16 of these had a frequency of ≥ 1 per cent and were thus sampled for the field measurements. The remaining 27 sub-clusters corresponded to 1755 grid cells, i.e. only 3.8 per cent of the forested area. According to the distribution of the field-measured cells in terms of the H_{max} and CC features (Figure 1) and their spread over the forested area (Figure 2), the sampled plots are expected to be a good representation of the entire grid.

In the field, five plot locations were not measured due to observations that the location either was not on forest land or had been harvested after the ALS data acquisition. Additionally, 11 plots were rejected in the post-processing phase due to a difference between the observed and planned plot position being > 3 m. The remaining 158 plots were retained for the analyses. The difference between the observed and planned positions of these plots was found to be 1.2 m, on average, and < 1 m on altogether 103 plots. According to basal area proportions derived from the field

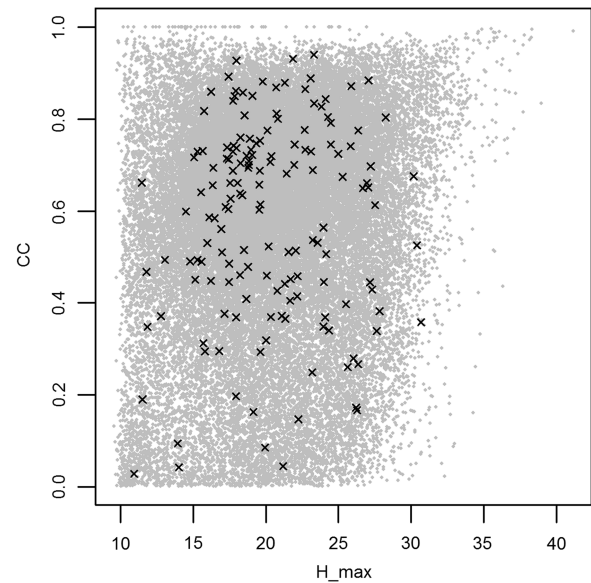


Figure 1 The distribution of the field-measured (black crosses) and all grid (gray dots) cells in terms of the average height of the highest 10% echoes (H_{max}) and the ratio of the echoes reflected from the canopy to all echoes (CC).

measurements, altogether 96 (61 per cent), 42 (26 per cent) and 20 (13 per cent) of the plots were dominated by pine, spruce and deciduous species, respectively. In total, 35 and 7 plots had a proportion of ≥ 95 per cent of Scots pine and Norway spruce, respectively, whereas purely deciduous plots were not observed in the field. The size and age distribution of the trees were overall found to correspond well with the characteristics expected for the area. For example, the ages of the median trees range from ~ 10 to 150 years for all the tree species groups considered, but most frequently ~ 50 years due to the silviculture applied in the area.

The 16 sub-clusters studied were labelled as A–P approximately according to an increasing H_{max} . The analysis on the persistence of the clusters showed an interesting pattern (Figure 3) that could be linked with the ALS features used for the clustering (Figure 4). By analysing the magnitude by which the cells persisted in the same cluster or shifted between the clusters, the four initial clusters using $k = 4$ and the clusters that were subsequently split from those could be identified (Figure 3). For example, the multiple interconnections between the sub-clusters with labels D–M indicate that these clusters were initially and also frequently after the first split occurring in the same clusters, but were separated when the partitioning was extended. Consequently, these classes may be expected to be more related in terms of the structure, whereas classes A and B and N–P are assumed to form more distinct structural groups. The deviations of the class-specific mean values from the population-level mean values (Figure 4) indicated differences in terms of the forest structural metrics between these classes, which were verified by the field measurements.

When examined against the field measurements, the clustering was somewhat related to forest size attributes (Figure 5), but not to those related to the understorey, shrub and herb layers (Figure 6). Particularly, the aggregated stem volume and basal area (Figure 5, upper row) showed patterns that could be linked with the distribution of the sub-clusters according to the ALS features (Figure 4).



Figure 2 The spatial distribution of the forest and the field plots in the studied area.

Based on the field measurements (Figure 5), sub-cluster A included the most of the youngest stands, but was not uniform according to the considerable variation in the diameters. The maturity and stocking increased according to the cluster labels in B–E and B–J in terms of height and basal area or volume, respectively. Sub-clusters I and J were the most intensively stocked. Sub-clusters K–O exhibited a similar but opposite pattern in terms of the tree stocking than B–J (Figure 5, upper row). The difference is that K–O had up to two vertical strata, whereas B–J were predisposed to several vertical strata (Figure 6, upper right). Sub-cluster O included the most variation in the diameter distribution. Only sub-cluster P could not be characterized according to these measurements.

The development stages corresponding to the reasoning of the previous paragraph were not recorded in the field. However, class P was remarked to generally consist mainly of seedling stands or other sparse tree positions such as stand boundaries and driving paths. Class O was remarked as matured forest, which would have posed an ideal habitat for the bird species considered except for the applied even-aged structure, i.e. removed understorey. The initial clusters or those split early in the clustering hierarchy (Figure 3) were found to correspond to more mature,

large and sparse forest stands than the representatives of other sub-clusters.

Since the grade given in the field was a direct observation of the habitat quality, it was interesting to examine whether any sub-cluster was particularly related to it. However, like in the case of the other statistics on the understorey, shrub and herb layer properties, no clear relationship was found (Figure 6, upper left). For example, the highest average and individual grades were found from the sub-cluster K which, however, also included plots with very low grades. Therefore, picking grid cells represented only by sub-cluster K as the areas prioritized for habitat management would have resulted in both an omission error due to not finding all potential habitats located in the other sub-clusters and a commission error due to including unsuitable habitats based on the field observations.

Based on a comparison between field data and the habitat requirements reported in the literature, plots that had been marked with a grade ≥ 9 (altogether six plots) or ≥ 7 (five plots) were determined as the most potential habitats for *T. urogallus* or *Tetrastes bonasia*, respectively. These plots were found rather distinct in terms of the ALS features, when the five features were

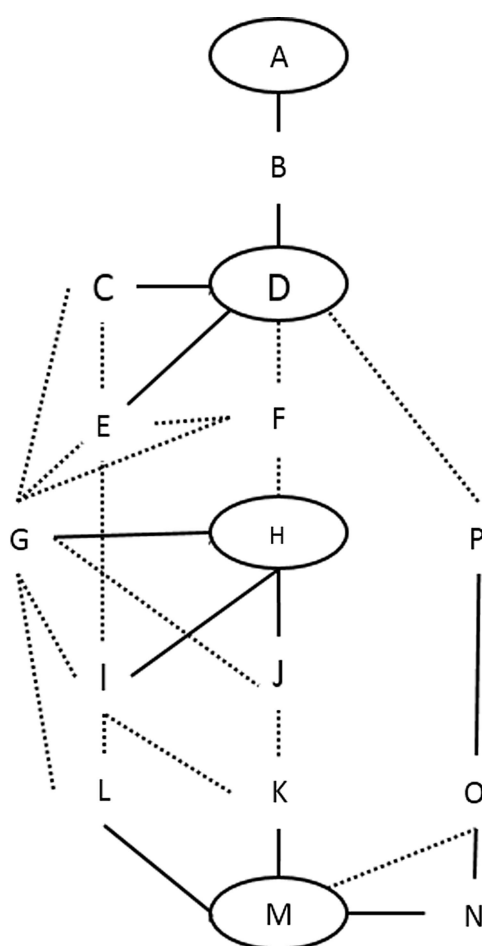


Figure 3 The persistence of the 16 field-measured sub-clusters in the clustering. The clusters circled by ellipses did not change during the clustering, while the lines show how the others were inter-connected. The solid and broken lines indicate that the classes belonged to the same cluster in ≥ 50 and $< 50\%$ of the partitions, respectively. The labels are assigned approximately according to an increasing average height of the highest 10% echoes; otherwise, the positions of the classes in the diagram are arbitrary.

jointly evaluated (Figure 7). Attempting to detect *T. urogallus* habitats according to the ranges of the ALS metrics on the identified plots yielded altogether nine plots. This selection included five out of six plots graded ≥ 9 for this species and four plots with a lower grade, which corresponds to omission and commission error rates of 17 and 44 per cent, respectively. A corresponding selection of the *Tetrastes bonasia* marked plots with a grade ≥ 7 yielded six plots, an omission error of 20 per cent and a commission error of 33 per cent.

By applying the criteria presented in the previous paragraph, altogether 1009 and 912 cells were selected from the full grid as potential *T. urogallus* (grade ≥ 9) and *Tetrastes bonasia* (grade ≥ 7) habitats, respectively, corresponding individually ~ 2 per cent or altogether 4.2 per cent of the forested area. The selected *T. urogallus* habitats composed of larger connected areas, whereas the targets expected suitable for *Tetrastes bonasia* were often located in separate, ribbon-shaped forms around stand borders (Supplementary

Figure, top). Both of these spatial aggregations are realistic according to the real-world occurrence of the considered birds. Interestingly, the initial clustering formed stand-like mosaicking patterns (Supplementary Figure, bottom), even though the neighbourhoods of the grid cells were not considered, but the classification was based on the ALS data of the individual cells.

Discussion and conclusions

According to our results, obtaining a detailed characterization of the habitat quality for the species considered here is challenging based on ALS data alone. The information obtainable with respect to the vegetation below the dominant canopy in particular is rather limited. Yet, the shrub and herb layers constitute important habitat characteristics due to determining the availability of nutrition and shelter for the birds. Further breeding and courting season-specific differences in the habitat selection were not even considered. Nevertheless, the description of structural differences in the overstorey based on ALS data is extremely useful information for prioritizing certain forest areas for habitat management.

The unsupervised classification of the ALS data stratified the forest area in terms of the size-related attributes (Figure 5) in particular and formed aggregated, stand-like spatial patterns, even though the neighbourhoods of the individual grid cells were not considered. The *a priori* information on the forest structural differences obtained by such a pre-stratification could be readily useful for forest stand delineation based on multi-scale information (Tokola *et al.*, 2008; Wu *et al.*, 2014), designing field plot sampling (Dalponte *et al.*, 2011; Maltamo *et al.*, 2011; Gobakken *et al.*, 2013) or applications requiring pre-stratification in general (see also Maltamo *et al.*, 2015). Practitioners are readily able to adopt the proposed approach in an operational workflow, since producing the unsupervised classification itself does not require any additions to the data already collected for the aforementioned applications. Whether there are resources available to establish a control inventory, as described above, further links between the data-driven clustering and real-world phenomena such as habitat characteristics may be discovered.

Even though the first examples of describing bird habitat structures based on ALS data were carried out a decade before the present study (Hill *et al.*, 2004), the first studies in the Scandinavian boreal forest structures corresponding to our study area are considerably more recent (Eldegard *et al.*, 2014; Lindberg *et al.*, 2015; Vihervaara *et al.*, 2015). However, these studies did not consider *T. urogallus* or *Tetrastes bonasia*, and with respect to these species, the closest examples are those by Central European researchers (Breidenbach and Braunisch, 2008; Graf *et al.* 2009; Zellweger *et al.*, 2013, 2014). The differences between our approach and the earlier studies can be structured according to the modelling approach, the predictor features and the scale used.

Unlike previous studies, our approach was based on an unsupervised classification of the data. The habitat requirements of the bird species considered were identified according to the previous literature, but such classes could not be produced when the data were partitioned in an unsupervised mode. Evidently, supervised methods should be favoured in the case of habitat mapping of the studied species. Labelling the unsupervised result with data obtained by a minor field check, however, indicated a possibility to improve the success rates of the clustering. Applying an

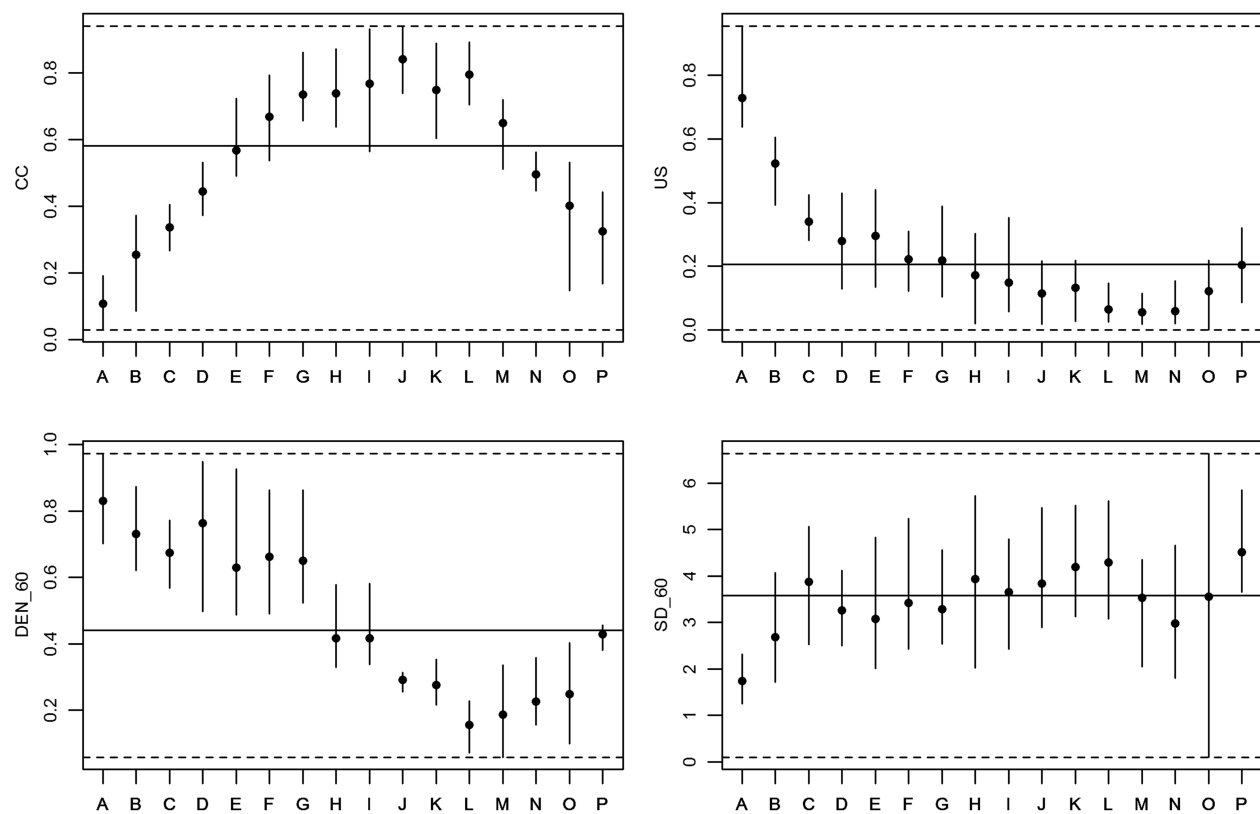


Figure 4 The distribution of the ALS features (the ratio of the echoes reflected from the canopy to all echoes – CC, the ratio of the echoes reflected above ground but below 60% of the maximum height to all echoes reflected above ground – DEN_60, the standard deviation of the height values of echoes reflected above ground but below 60% of the maximum height – SD_60 and the ratio of the echoes reflected from the shrub layer to all echoes – US) according to the sub-cluster labels of Figure 3 arranged approximately according to an increasing average height of the highest 10% echoes. The black dots indicate the cluster-specific mean values and the vertical lines the range of the values within the clusters. The horizontal solid and broken lines indicate the population-specific mean and extreme values, respectively.

unsupervised classification for selecting the observations to be measured for training data or a purely semi-supervised labelling approach should be further studied with respect to optimally selecting and utilizing the field sample in the classification phase (see, e.g. Theodoridis and Koutroumbas, 2009, pp. 577–590). According to the selection based on the semi-supervised regulation, a proportion of 4 per cent of the forested area was proposed suitable as habitats of the considered species. However, this figure may be subject to similar omission and commission error rates as those reported based on the training sample, which could not be verified using an independent validation. It should also be interpreted as the proportion of the core area of these habitats, i.e. a perfect suitability of the 254–256 m² computation units. The ‘true’ habitats of these species are likely composed of such core areas and the surrounding forests forming territories of 15–25 ha (*Tetrastes bonasia*) to up to 300 ha (*T. urogallus*).

The *k*-means approach is based on partitioning the feature space to a user-defined number of clusters. There is no objective criteria for determining the number of clusters, and this decision has earlier been based on experimental or theoretical *a priori* knowledge over various cluster solutions (Pascual et al., 2008, 2013). By initially experimenting with different numbers of clusters, we found that four to seven structural classes could be separated based on our study area and data. For example, site types and

development stages are both based on five classes applying to the studied forest structures. However, not all classes are likely present in the study area and, on the other hand, the ability of the data to describe all these properties is limited. Increasing *k* significantly from the aforementioned values resulted in clusters representing a very low number of grid cells, which did not reflect the true extent of the habitats studied. However, the selection of initial *k* = 4–7 is subjective, but we believe that its effects were somewhat alleviated by analysing the persistence of the clusters.

In the final clustering approach, the solutions based on the various clusters were examined to determine the persistence and inter-connections of the observations during the clustering. The approach somewhat resembles the ISODATA or ISO-clustering (Jensen, 1996), which is a standard technique for partitioning remote sensing data with applications in land use mapping, for example. In the latter, the requirement to determine the number of clusters *a priori* is avoided by split and merge criteria based on the occurrence or diversity of the clusters obtained, whereas in our approach, this decision is based on more detailed information on the clustering hierarchy.

The selection of ALS features was based on the available literature about the habitat requirements of the targeted bird species. The predictor features slightly differed from previous studies, but were expected to fundamentally depict the same

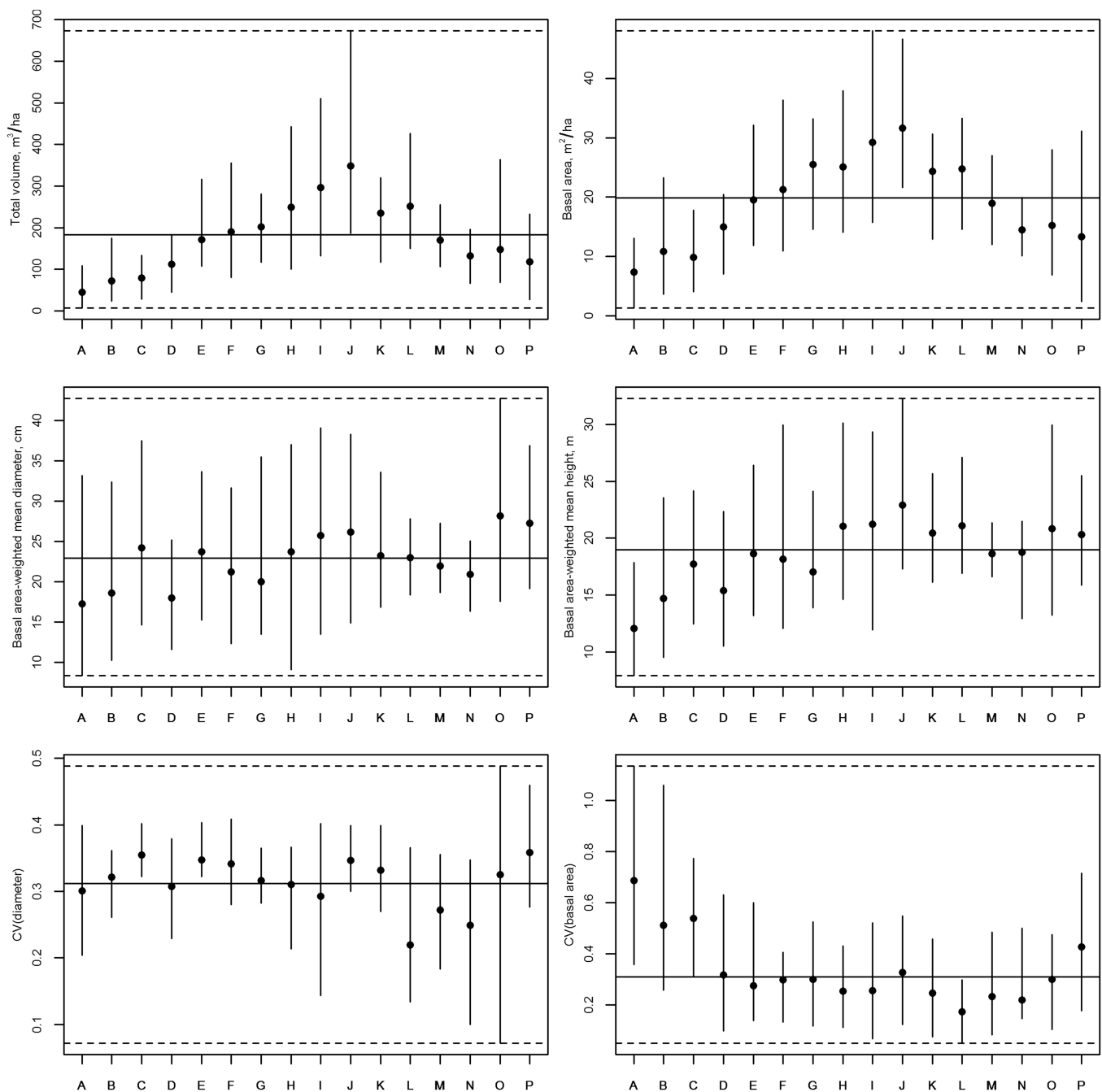


Figure 5 The distribution of the field-measured forest structural attributes according to the sub-clusters. CV(diameter) and CV(basal area) refer to the coefficient of variation of the diameters within the fixed plots and the coefficient of variation of the basal areas between the satellite plots, respectively. For the interpretation of the cluster labels on the x-axis and the symbols used, see Figures 3 and 4.

forest phenomena as quantified by Breidenbach and Braunisch (2008), Graf *et al.* (2009) and Zellweger *et al.* (2013, 2014) using ALS data with closely corresponding acquisition parameters. In addition to these studies, further research on the quantifications of forest structure by ALS (e.g. Zimble *et al.*, 2003; Frazer *et al.*, 2005; Maltamo *et al.*, 2005; Korhonen *et al.*, 2011) suggest that the vertical heterogeneity, horizontal canopy cover and the distribution of patches are fairly well

assessed and related to different types of forest across the boreal and temperate biomes. However, there are well-known limitations in linking the within-patch information to the actual tree pattern (Packalén *et al.*, 2013) and understorey (Korpela *et al.*, 2012). Even if indicative information on the understorey vegetation cover and density may be obtained (Goodwin *et al.*, 2007; Wing *et al.*, 2012), this information is prone to transmission losses due to the upper canopy (Korpela *et al.*, 2012) and

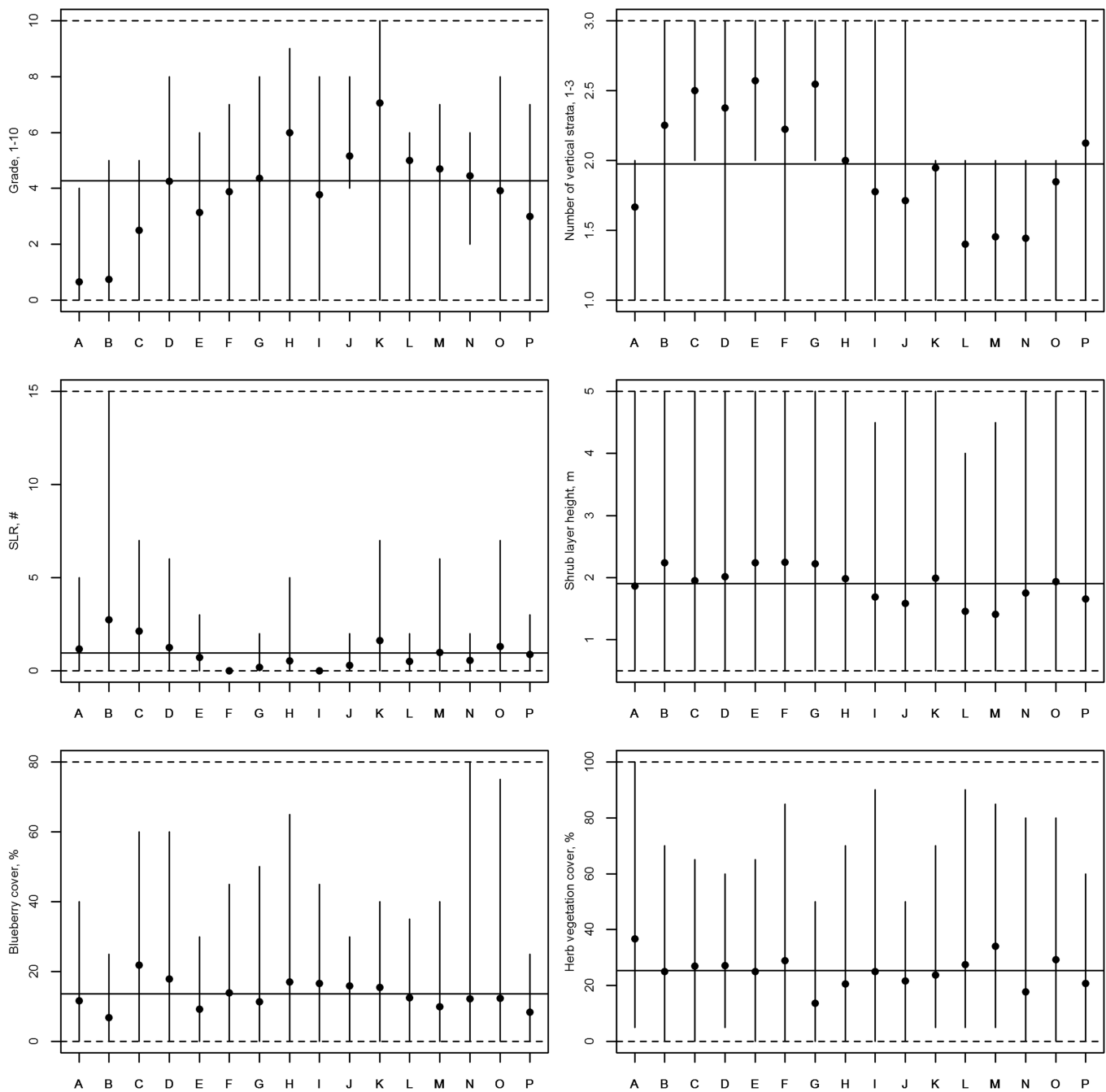


Figure 6 The distribution of the field-measured understorey and shrub layer properties according to the sub-clusters (SLR – number of spruces with low branches). For the interpretation of the cluster labels on the x-axis and the symbols used, see Figures 3 and 4.

therefore limited in forests with high canopy cover in particular (see also [Maltamo et al., 2005](#)).

It should be noted that in addition to the metrics typically used for studying habitat structure, ALS allows the derivation of numerous other predictor features to be considered for different types of applications. For example, predictors related to the understorey ([Bouvier et al., 2015](#)), canopy gaps ([Vehmas et al., 2011](#)) or a detailed volumetric characterization of the vertical strata ([Lindberg et al., 2012](#)) could be considered. Even a simple quantification of

the canopy surface texture could reveal below-surface structural variation, as suggested by [Ozdemir and Donoghue \(2013\)](#) and [Wood et al. \(2012\)](#) based on ALS data and aerial images, respectively. However, computing these predictors is often permitted only by very high pulse densities, waveform-recording sensors or other data sources as opposed to the discrete-return data of $0.7\text{--}0.8\text{ pulses m}^{-2}$ used here. We acknowledge that a more detailed parameterization of the analysed landscape could have improved the results. However, requiring more detailed data

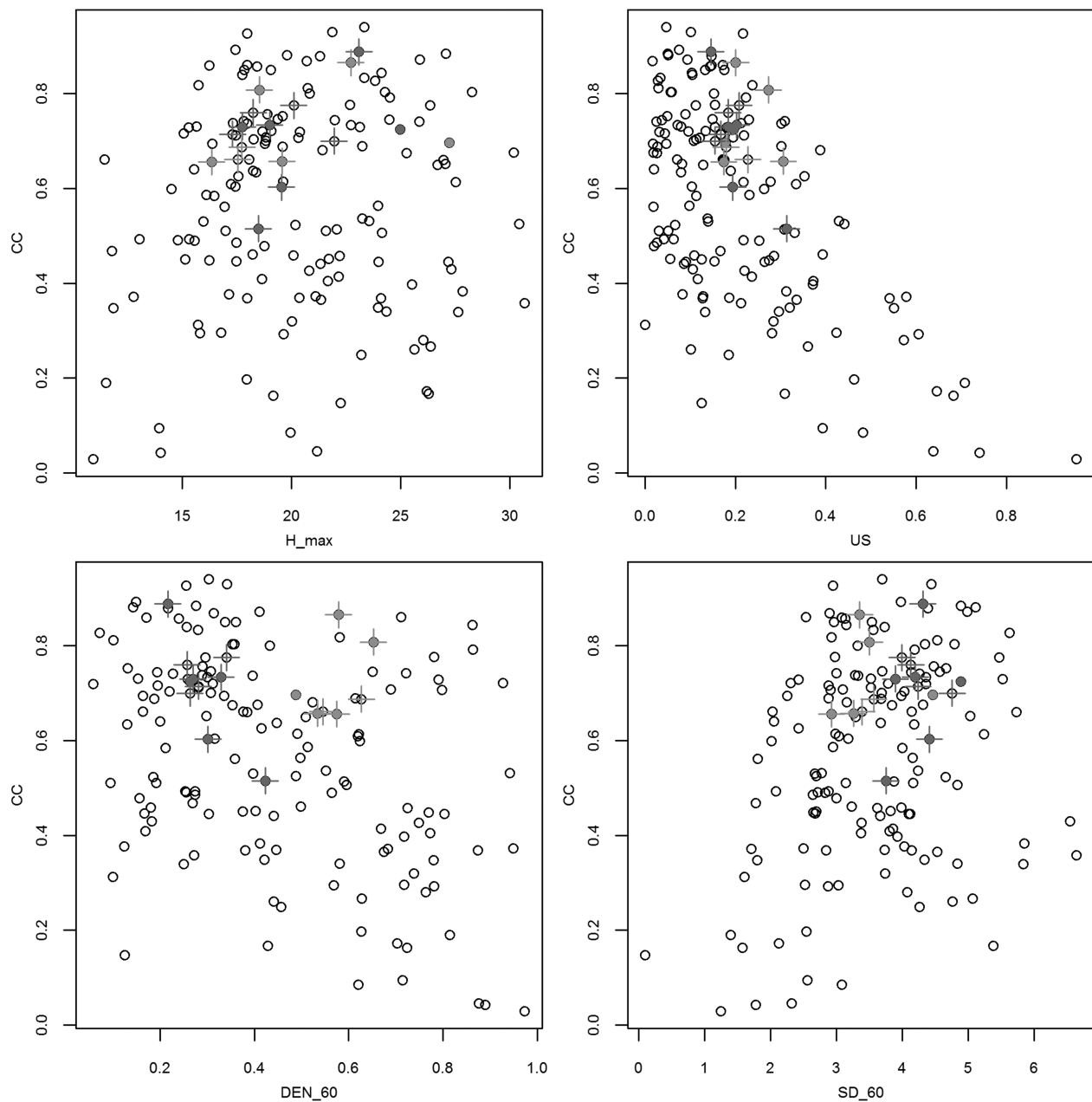


Figure 7 The most potential *T. urogallus* or *Tetrastes bonasia* habitats according to the field data highlighted for ALS feature values. The black circles represent the observations from the field-measured plots. The filled circles and crosses correspond to observed habitats and those selected according to the range of the observed habitats, while dark and light grey filling corresponds to *T. urogallus* and *Tetrastes bonasia*, respectively. H_{max} – the average height of the highest 10% echoes, CC – the ratio of the echoes reflected from the canopy to all echoes, DEN_60 – the ratio of the echoes reflected above ground but below 60% of the maximum height to all echoes reflected above ground, SD_60 – the standard deviation of the height values of echoes reflected above ground but below 60% of the maximum height and US – the ratio of the echoes reflected from the shrub layer to all echoes.

would mean distinct acquisitions with considerably higher inventory costs, whereas our approach can be operated based on ALS data, which already cover broad areas and are readily available for practical inventories due to the frequent acquisitions of such data for ground elevation modelling.

Finally, a considerable difference to the earlier studies, particularly those by Graf et al. (2009) and Zellweger et al. (2013, 2014), is the scale analysed. Graf et al. (2009) extracted the predictor

features using scales of 125×125 m and 500×500 m. In Zellweger et al. (2014), the predictor features were extracted from slightly smaller (10×10 m) computation units than here, but the results were analysed in a scale of 1 km^2 , in which the reference observations were collected. Aggregating at similar landscape levels could add the present pixel-level information on aspects such as the connectivity between forest patches and the spread of the habitats. The aforementioned scales are not suitable, however, considering

forest management planning decisions on compartments typically 0.5–5 ha and on average 2.5 ha in size, which need to be assigned either to timber production or to non-commercial uses such as maintaining habitats. Whether a multilayered stand is intensively thinned or treated by less intense management practices to preserve the heterogeneity will considerably affect the suitability of the stand as a habitat for certain species. Generating prioritization maps on the potentially existing habitats clearly requires approaches based on small initial computation units. Spatial optimization methods readily exist to aggregate units with similar properties into larger regions (Bettinger *et al.*, 2002; Strange *et al.*, 2002; Heinonen and Pukkala, 2007) and an application of prioritizing areas for the management of *T. urogallus* lekking sites based on forest inventory attributes obtained from coarse satellite image data is described by Sirkiä *et al.* (2012).

Overall, the majority of the ecological applications of ALS are currently focussed on assessing the habitat structure and species diversity explicitly for nature conservation planning. Such assessments may not directly serve forest management planning, where the aim is often to allow parallel rather than exclusive objectives. The methods proposed here produce prioritizations of habitats, which could be further examined at aggregated or multiple scales. Similar maps could potentially be generated to prioritize various other ecosystem services. Overlaying such maps using expert opinion (e.g. Kangas *et al.*, 2000; Pascual *et al.*, 2013) could further progress multi-criteria decision analyses (Kangas *et al.*, 2008) for the management planning of at least a proportion of the forest area considered. Our future research interests are related to this direction.

Supplementary data

Supplementary data are available at *Forestry* online.

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Conflicts of interest statement

None declared.

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